Grab the slides

NEXTUP SOCAL · VIRTUAL EVENT

Reducing AI Operating Costs

What the tokens actually cost, why the invoice grows faster than the usage, and the five levers that bring it back down.

Nick McCarty

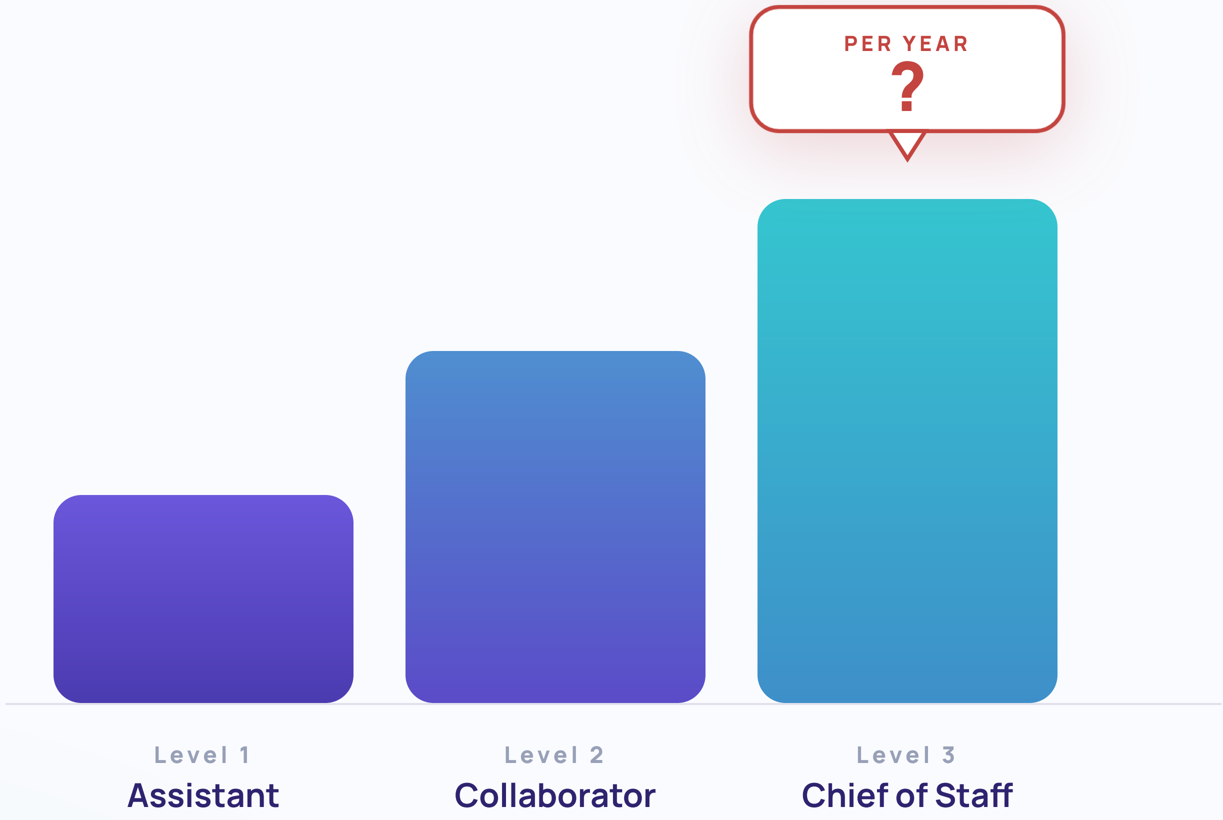
Upskilled Consulting

Fri Aug 21, 2026 · 12:00 PM PST

Part two of the NextUp SoCal AI series

WHERE WE LEFT OFF · JUNE 5

In June we mapped the levels. Today: what each one costs to run.



- **Level 1 — Assistant.** One prompt, one answer. Cheap enough that nobody checks.
- **Level 2 — Collaborator.** The tool knows your context. More reading, more tokens.
- **Level 3 — Chief of Staff.** It works without you. It reads *constantly*.

WHERE TODAY PICKS UP

74% of companies plan to deploy agentic AI within two years. So the useful question now is a budgeting one: what does Level 3 cost to run for a year?

Adoption figure from the June 5, 2026 NextUp SoCal session (Deloitte).

THE HOUR

From “it’s cheap” to a number you can budget.

01 · THE UNIT

What is a token, and what does one actually cost?

Pages, meetings, emails — converted into the thing you’re billed for.

By the end of it you can price any document your team touches.

02 · THE MULTIPLIER

Why a fraction of a cent becomes half a million dollars.

Three scenarios, same 600 people, same price list, 47× apart.

Same tool, same headcount — three very different invoices.

03 · THE LEVERS

Five ways to cut the bill without cutting the capability.

- Right-size the model
- Script the plain steps
- Engineer the context
- Stop redoing work
- Run commodity work local

What you don't spend is also a lever

What you keep out of the prompt, and what you keep out of your stack.

And the cost of standing still

The most expensive line item is usually the work you didn’t automate — and who in the room never got the tool.

01 · THE UNIT

You’re not billed per prompt. You’re billed **per token**.

THE QUESTION

`Sum` `mar` `ize` `the` `Q` `3` `category` `review` `deck` `for` `the` `leadership` `team` `.`

THE ANSWER

Category volume rose 4.1% against a 2.3% plan, but nearly all of the gain came from promotional units — base velocity was flat. Gross margin fell 60 basis points, driven by deeper promo depth in weeks 7 through 11 and a mix shift toward the value tier. Two SKUs account for 71% of the margin decline; both ran continuous promotion all quarter. Recommend holding promo depth flat through the December reset, moving those two SKUs to an every-other-week cadence, and revisiting value-tier facings at the January line review.

TOKENS IN	TOKENS OUT	COST OF THE QUESTION	COST OF THE ANSWER
14	**180**	**$0.00002**	**$0.0018**
the question you asked	the summary it wrote back	14 tokens @ $1.25 / 1M	180 tokens @ $10 / 1M

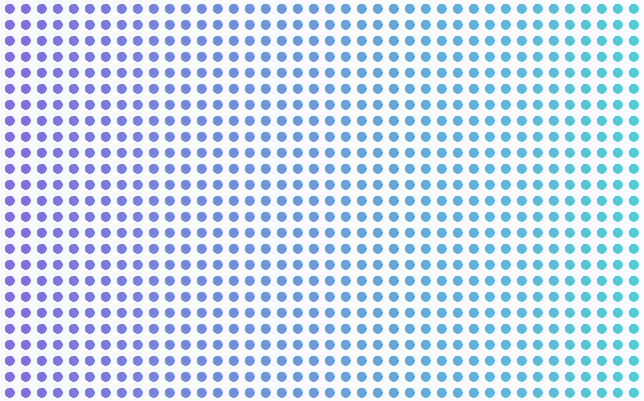
Output is priced **8× higher** than input — $10 vs $1.25 per million tokens. On this request **99% of the cost is the answer**. Agents flip that completely: they read far more than they write.

Token counts approximate; every model tokenizes slightly differently. Prices: GPT-5 standard tier, $1.25 / 1M input, $10 / 1M output.

Your work, converted into the billable unit.

Every artifact your team touches has a token weight.

ONE THOUSAND TOKENS



≈ 750 words

≈ 4,000 characters

≈ 1½ pages of text

WHAT YOUR ARTIFACTS WEIGH, IN TOKENS

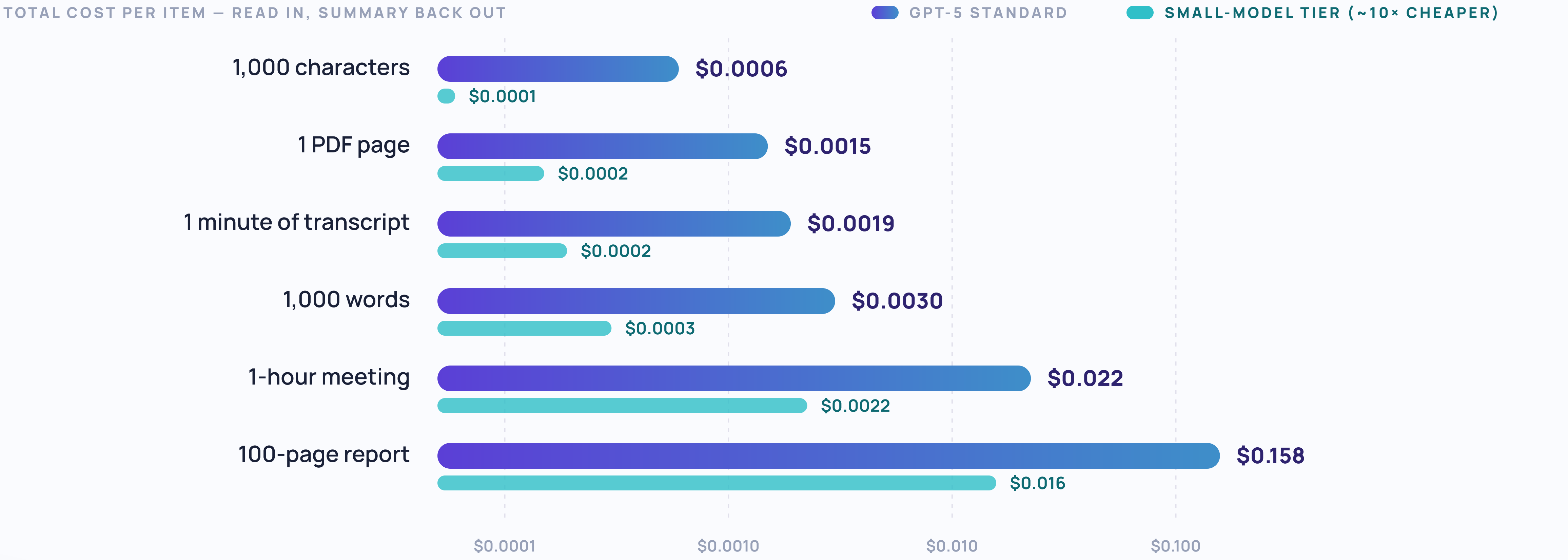
70× the meeting — and, as you're about to see, still under twenty cents.

Speech figure assumes 130–160 wpm. Image tokens ≈ (width × height) ÷ 750 for common vision models.

01 · THE UNIT

Here is what **one of everything** costs.

Read it in, hand a summary back. Common artifacts, priced.



Nothing here is expensive — which is why the invoice often surprises people as they move up the levels.

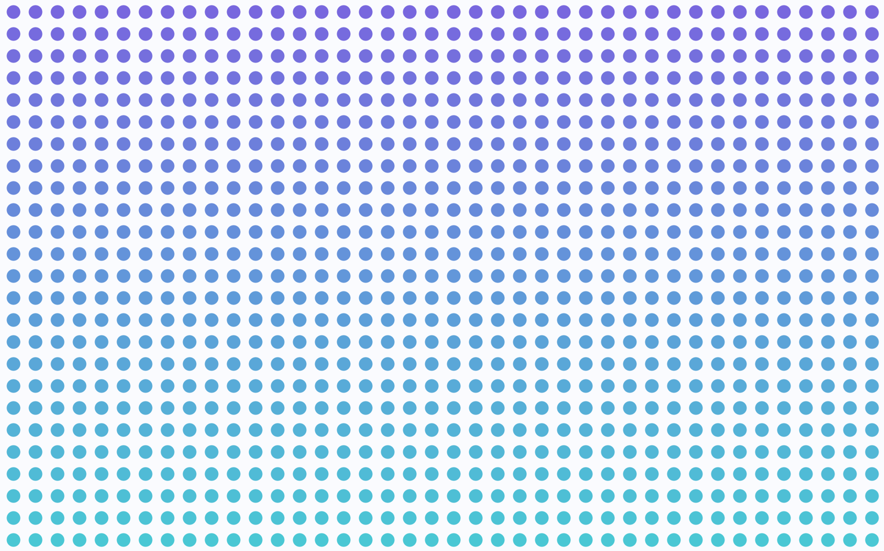
GPT-5 standard tier ($1.25 / $10 per 1M). Summary assumed at 10% of input. Small-model tier drawn at 1/10 the token price.

01 · THE UNIT

For perspective: what everyday work costs.

A single email	< $0.001
A 10-page report	≈ 2¢
A one-hour meeting	≈ 2¢
A 100-page report	≈ 16¢
A day of email — 50 messages	≈ 5¢
1,000 reports × 100 pages each	≈ $158

1,000 REPORTS · 100 PAGES EACH



$157.50

READ AND SUMMARIZED, FOR ABOUT THE PRICE OF ONE TEAM LUNCH

The unit is cheap. The bill is a volume problem — and volume is a decision we have influence over.

Derived from the per-artifact costs on the previous slide: input tokens plus a summary at 10% of input length.

02 · THE MULTIPLIER

Nobody's budget breaks on price. It breaks on volume.

$0.02	×	40	×	600	×	250	=	$120,000
COST PER TASK		TASKS PER DAY		PEOPLE		WORKING DAYS		A YEAR

The assistant. Chat, summarize, draft.

PER EMPLOYEE, PER DAY

INPUT — WHAT IT READS

20,000

tokens

OUTPUT — WHAT IT WRITES

5,000

tokens

AT GPT-5 STANDARD PRICING

$1.25 per 1M input tokens

$10 per 1M output tokens

WHERE THE DAILY COST GOES

input $0.025 · 33%

output $0.050 · 67%

DAILY, PER EMPLOYEE

$0.075

× 600 employees × 250 working days

EVERY YEAR

600 employees × 250 working days. GPT-5 standard tier: $1.25 / 1M input, $10 / 1M output.

SCENARIO 2 OF 3 · LEVELS 2–3

The agent. Agents read far more than they write.

PER EMPLOYEE, PER DAY

INPUT — WHAT IT READS

500,000

tokens

OUTPUT — WHAT IT WRITES

20,000

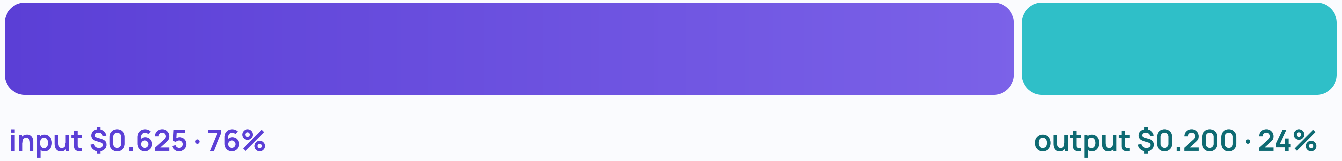
tokens

AT GPT-5 STANDARD PRICING

$1.25 per 1M input tokens

$10 per 1M output tokens

WHERE THE DAILY COST GOES



DAILY, PER EMPLOYEE

$0.825

× 600 employees × 250 working days

EVERY YEAR

11× scenario 1 — same 600 people

600 employees × 250 working days. GPT-5 standard tier: $1.25 / 1M input, $10 / 1M output.

SCENARIO 3 OF 3 · PAST LEVEL 3

Always-on automation. Agents that never stop watching.

PER EMPLOYEE, PER DAY

INPUT — WHAT IT READS

2,000,000

tokens

OUTPUT — WHAT IT WRITES

100,000

tokens

AT GPT-5 STANDARD PRICING

$1.25 per 1M input tokens

$10 per 1M output tokens

WHERE THE DAILY COST GOES

input $2.50 · 71%

output $1.00 · 29%

DAILY, PER EMPLOYEE

$3.50

× 600 employees × 250 working days

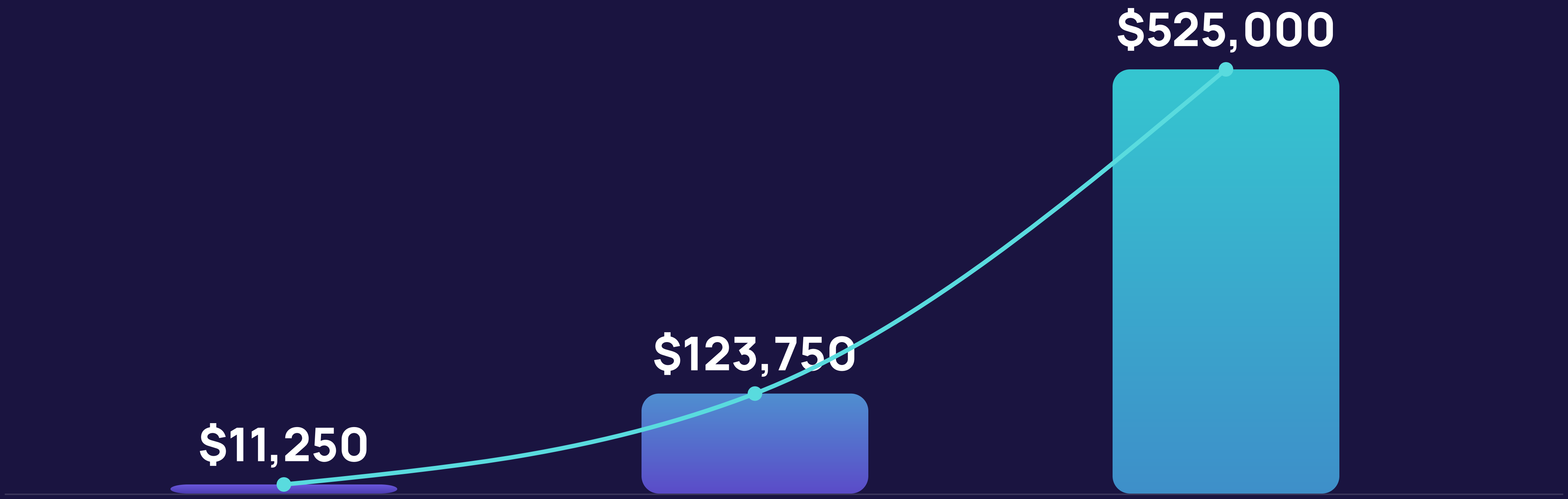
EVERY YEAR

47× scenario 1 — same 600 people

600 employees × 250 working days. GPT-5 standard tier: $1.25 / 1M input, $10 / 1M output.

02 · THE MULTIPLIER

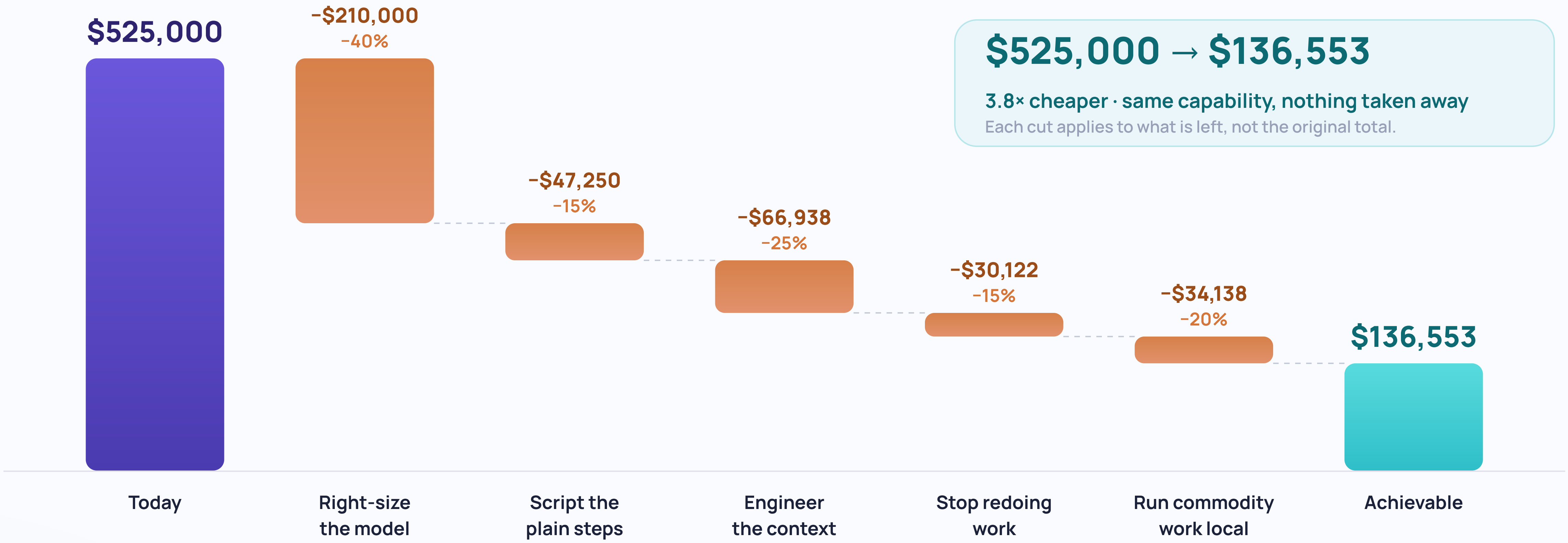
Same 600 people. Same 250 days. Same price list. 47× apart.



SCENARIO 1	SCENARIO 2	SCENARIO 3
Assistant	Agentic	Always-on agents
$18.75 per person / year	$206.25 per person / year	$875.00 per person / year

Annual inference cost at GPT-5 standard-tier pricing. Linear scale — scenario 1 is drawn to the same scale as the others.

Five levers. Same capability, a fraction of the bill.



$525,000 → $136,553

3.8× cheaper · same capability, nothing taken away

Each cut applies to what is left, not the original total.

Illustrative model. Each cut applies to the running remainder, not the original total. Your mix will differ.

LEVER 1 · RIGHT-SIZE THE MODEL

We're paying frontier prices for a 1.7% edge.

BENCHMARK SCORE — BEST PAID vs BEST OPEN-WEIGHT

Best paid model
premium per-token pricing

100

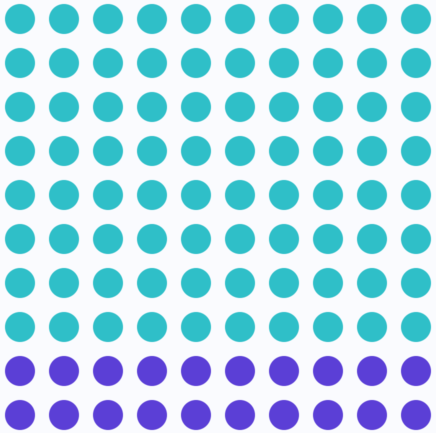
Best open-weight model
run it at a fraction of the cost

98.3

8% → 1.7%

performance gap, closed in a single year

OF 100 TASKS…



80 routine — cheap tier

20 genuinely hard — frontier

RELATIVE SPEND ON THE SAME 100 TASKS

Everything on the frontier model

100

Routed — 80 cheap / 20 frontier

28

$(20 \times 1.0) + (80 \times 0.1) = 28$

−72%

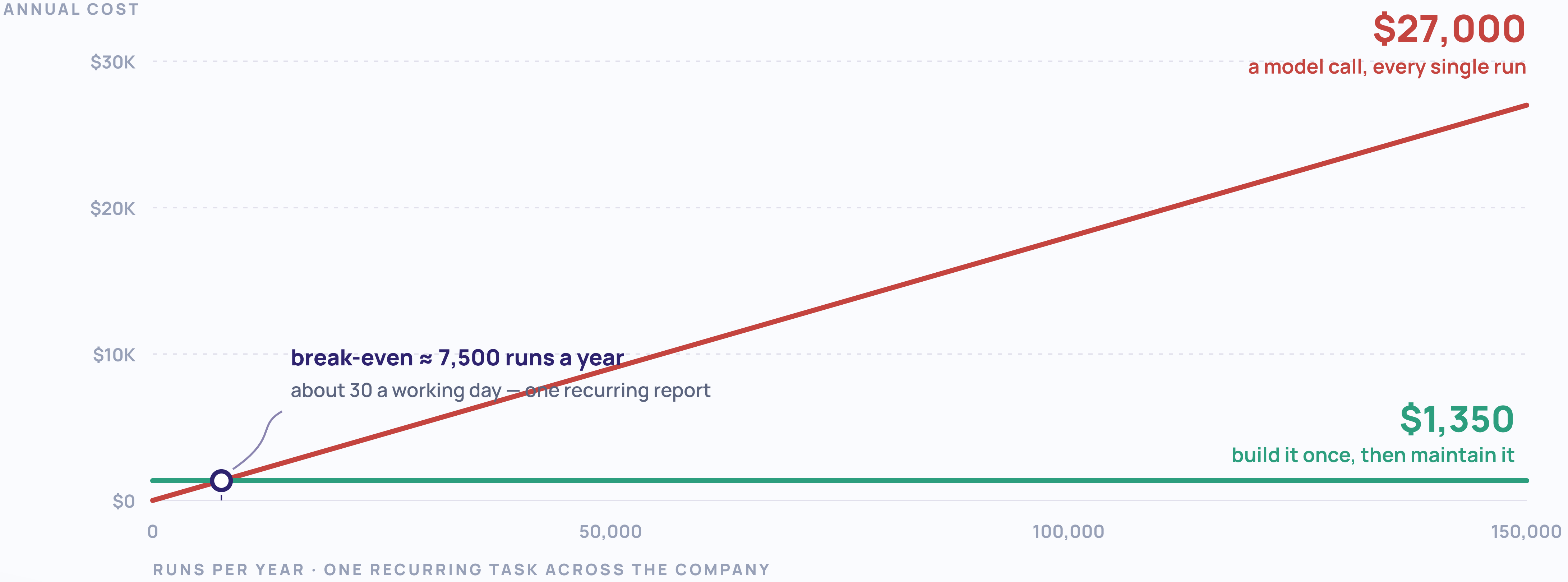
on the same work, from a routing rule

Defaulting to the premium model for every task is a tax on habit — not a quality decision.

Benchmark gap: Stanford HAI, 2025 AI Index Report. Routing math assumes the small tier is ~10× cheaper.

LEVER 2 · SCRIPT THE DETERMINISTIC STEPS

Most of what you're automating doesn't need to think.

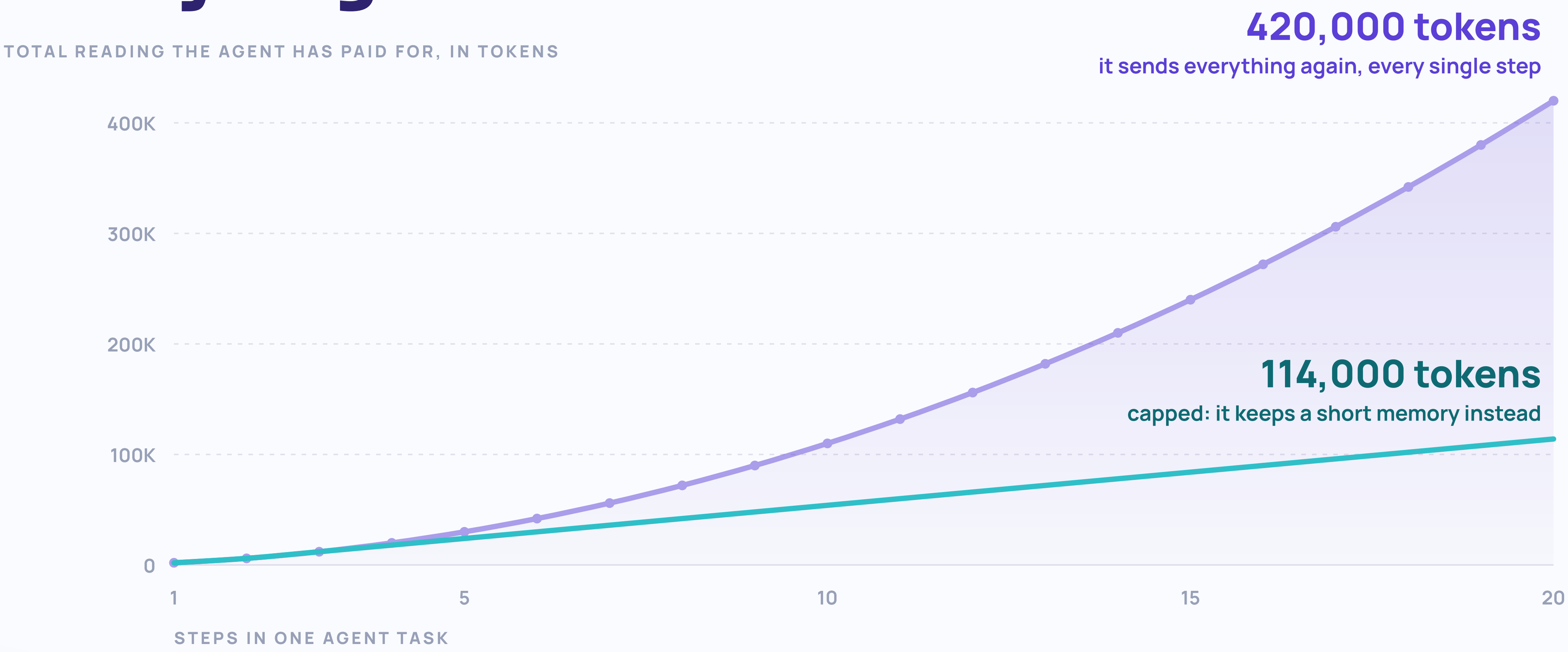


The test is simple: if you can write the rule down, you do not need to rent judgment to follow it.
Where the task genuinely needs judgment or language, the model earns its price. Spend it there.

Illustrative: $0.50 per agent run, a one-time build of ~$400 in analyst time, $1,200/yr maintenance.

LEVER 3 · ENGINEER THE CONTEXT

Your agent is re-reading the same page all day long.



Cap what the agent re-reads and the same work costs about a quarter as much.*

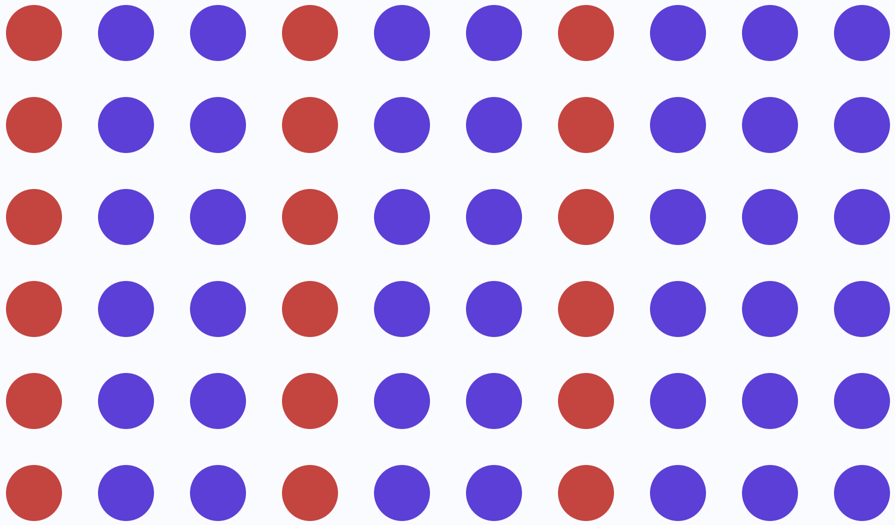
* Assuming two-thirds of tokens never reach the finished answer.

Illustrative: a 20-turn loop adding ~2,000 tokens per turn. Engineered = rolling window plus a running summary.

LEVER 4 · STOP REDOING WORK

The chat window is a shredder with a very good memory of nothing.

60 CHAT SESSIONS THIS MONTH



18 of them re-answered something already answered

and none of the answers were saved anywhere

AT 30% REWORK, ON A $525,000 BILL

$157,500

a year, re-answering answered questions

— before you count anyone's salary time

THE FIX IS FREE

Put the prompt and the output in version control.

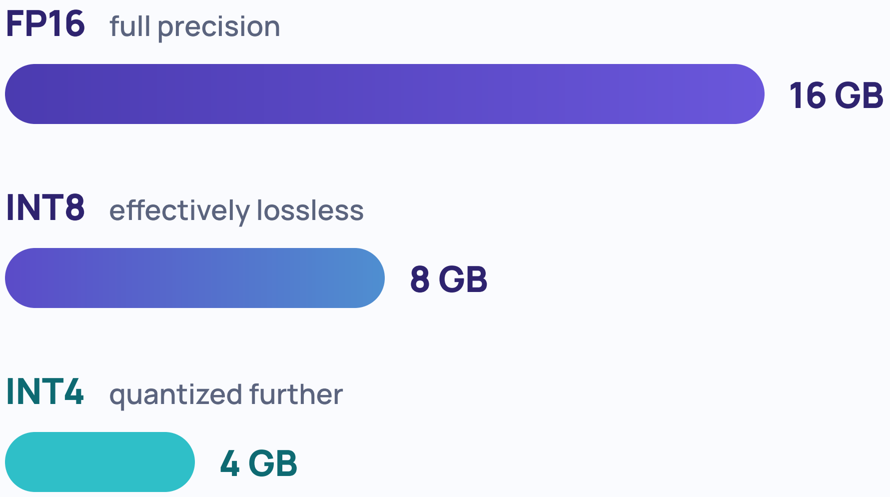
A prompt someone tuned for four hours is an asset.
So is the finished piece of work it produced.
Both are worth keeping where the team can find them.

Rework share shown at 30% — an assumption. Measure yours before budgeting against it.

LEVER 5 · RUN COMMODITY WORK LOCALLY

Smaller models, your hardware, nothing leaves the building.

AN OPEN-WEIGHT 8B MODEL FROM HUGGING FACE, IN MEMORY



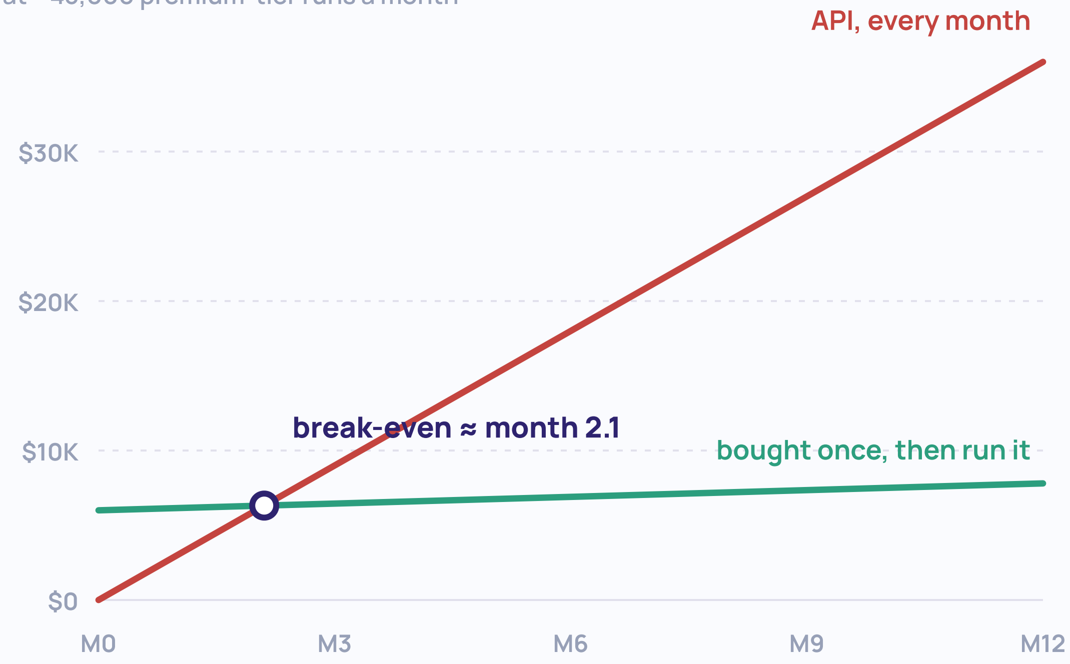
4 GB runs on a laptop.

Not a data center. Not a contract. A laptop.

Small enough that two or three specialists share one GPU: one for extraction, one for summarization, one for drafting.

CUMULATIVE COST — API vs YOUR OWN HARDWARE

at ~43,000 premium-tier runs a month



And nothing leaves the building.

* Local wins on volume — at commodity cloud pricing, electricity alone can cost more.

Memory = parameters × bytes per parameter (FP16 2, INT8 1, INT4 0.5). Break-even assumes premium-tier volume; it does not hold at commodity-tier pricing.

What you don't put in the prompt is a line on the budget too.

NEVER GOES IN

- API keys and passwords
- Customer names, emails, card data
- Client contracts and their IP
- Unreleased financials

FINE TO SEND

- The deck you're summarizing
- The meeting transcript
- Last quarter's public summary
- Your own draft email

It costs nothing to strip it out. It costs a great deal to explain it.

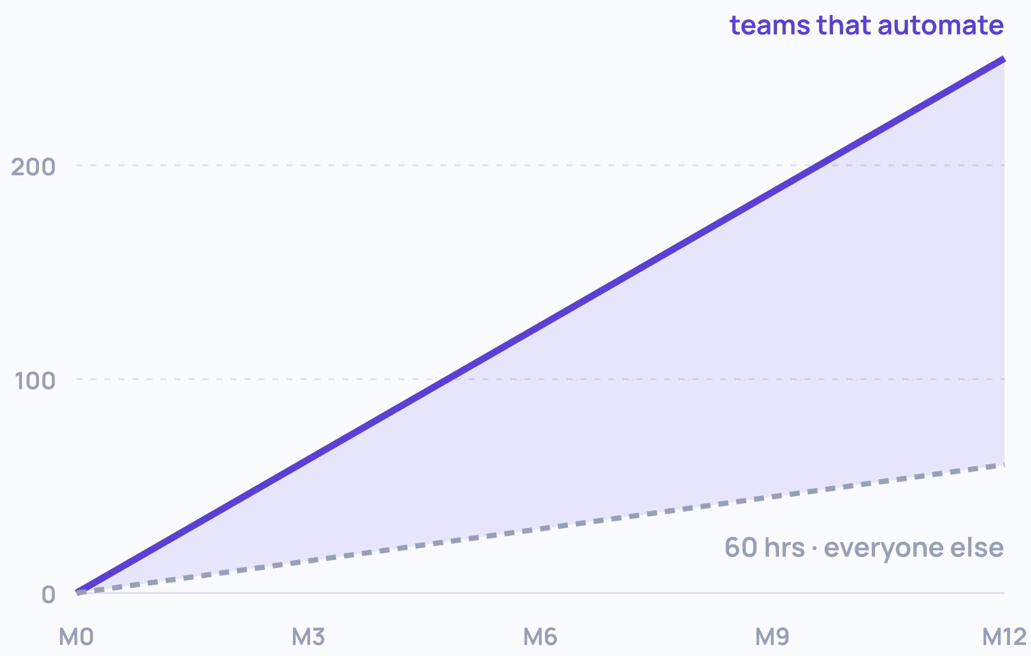
Fewer tokens in the prompt is also, literally, a smaller bill. Hygiene and thrift point the same way here.

Categories shown are the four that show up most often in enterprise prompt audits.

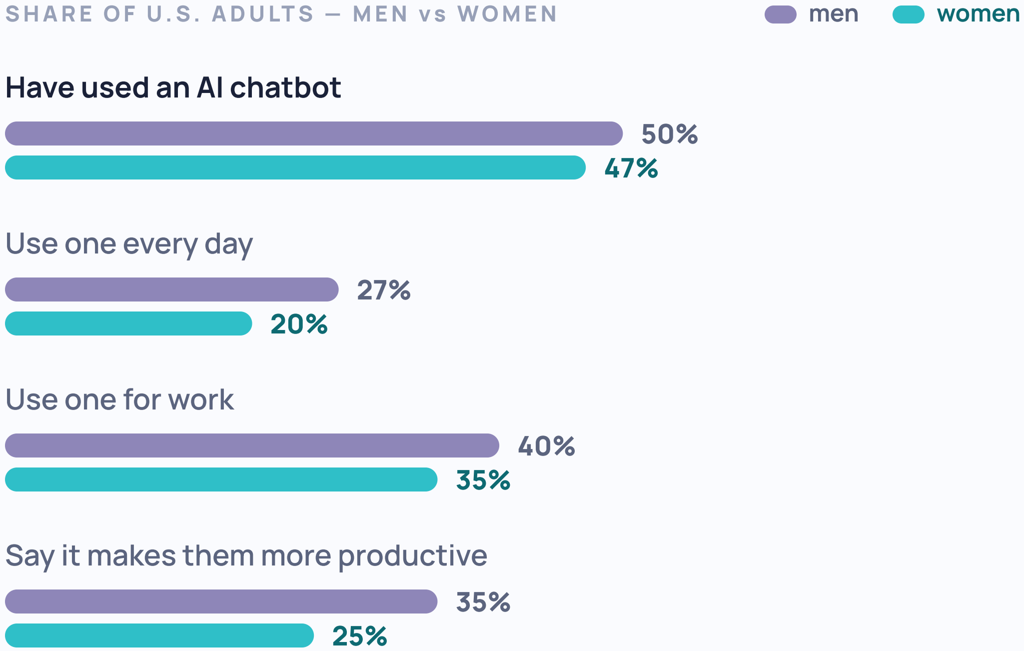
06 · WHO GETS THE LEVER

The most expensive line item is the one that isn't on the invoice.

HOURS FREED PER PERSON, CUMULATIVE



SHARE OF U.S. ADULTS — MEN vs WOMEN



The distance compounds.

> "The women who get ahead will be the ones using AI to multiply their output, not just speed up their typing." — Ashika Schroll

Pew Research Center, 5,119 U.S. adults, surveyed Feb 17–23 2026. Corroborated by Lean In's 2026 workplace survey (via Forbes).

The appetite is already there.
The training is the missing line item.

AI IS THE #1 TOPIC WOMEN SAY THEY WANT TO LEARN

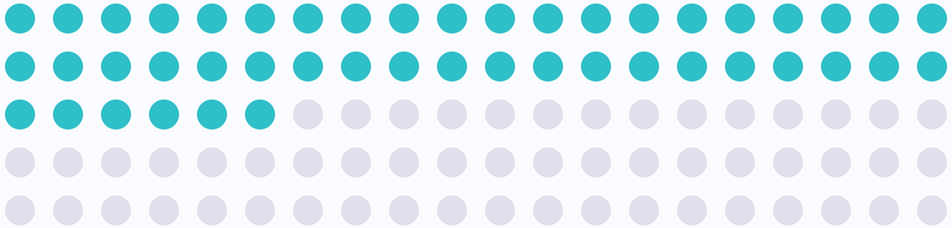
report a lack of skills and access to training on the job.

ACCESS TO AI TRAINING AT WORK

The constraint is not appetite. It is access.

WHAT THE C-SUITE ITSELF EXPECTS, WITHIN THREE YEARS

of workers will need reskilling because of AI.



Nearly half the people you already employ.

Every lever in this talk needs somebody who knows to pull it.
Routing, context and scripting do not happen because you bought a tool. They happen because someone was taught what to look for.

63%: Skillsoft Women in Tech Report. 46%: BCG, surveying C-suite expectations for reskilling over three years.

WHERE TO START

Four moves you can make without a budget request.

01 · METER IT

You cannot manage what you're not measuring.

Ask for cost per team, per month — and keep building your intuition for cost per workflow.

02 · ROUTE IT

Name your three biggest use cases. Work through two of them using the cheaper model.

Run the tasks on the cheaper model for two weeks and compare the output. If no material difference can be discerned, you've found savings.

03 · AUTOMATE ONE THING

Pick the task that runs the same way every week.

The Friday report nobody enjoys building. Ask for it to be automated once, and for the recipe to live somewhere the team can find it instead of one person's inbox.

04 · BUDGET PER OUTCOME

Set a price per thing, not per month.

A monthly cap tells you when you're in trouble; tracking cost per task tells you whether it was worth it.

LIVE DEMO

This is what “it reads constantly” looks like.

Video plays in the live deck only —
see **upskilled.consulting/nextup-socal**

25×

READS FOR EVERY WRITE

Reads (in) 500,000/day

Writes (out) 20,000/day

Same shape as Scenario 2, a few slides back. This is what that ratio looks like actually running.

ctarp, an agentic security scanner, localizing CVE-2024-2952 against the commit before its fix — no advisory, no hint, just tokens. Read/write ratio shown at left uses Scenario 2's figures.

WRAPPING UP

The takeaways worth keeping.

1. **Tokens are cheap. Volume drives the bill.**
   Summarizing a hundred-page report costs about a stamp — that alone won't break a budget. Agents that ingest your business artifacts all day and reason in circles about what to do with them can cost far more than anticipated.

2. **Agents read far more than they write.**
   As we demonstrated, always-on agents can easily cost 47x more than you're accustomed to paying for ad hoc chatbot sessions.

3. **Roughly three-quarters of cost is decision-driven, not a set price.**
   Routing, context and scripting get you most of the way back — without removing capabilities.

4. **Ask for cost per outcome, not cost per month.**
   Continue building intuition around cost per task and partnering across departments to begin analyzing monthly spend by task and workflow.

Questions?

Nick McCarty

Upskilled Consulting

nick@upskilled.consulting

linkedin.com/in/nicholasmccarty

Scan for the one-page cost checklist

the slides, and how to reach me.